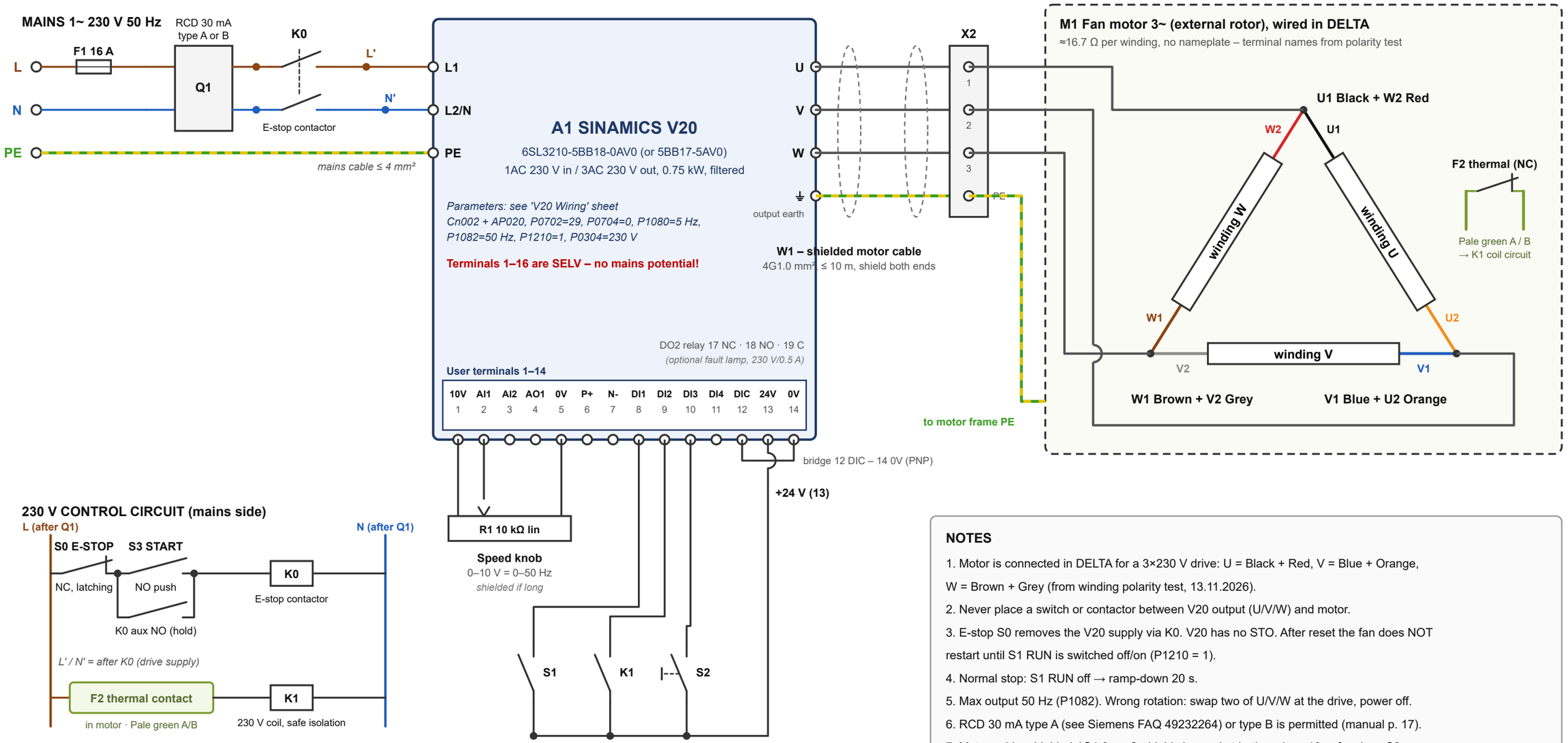
Wind-tunnel fan – SINAMICS V20 (1AC 230 V, 0.75 kW) to 3~ fan motor in DELTA

Wiring diagram for the electrician – DRAFT, to be verified and installed by authorised personnel. Terminal numbers per V20 Operating Instructions 09/2014 (p. 39–44, 66).



230 V CONTROL CIRCUIT (mains side)

K1 must have safe isolation coil/contact (≥ 4 kV). The thermal contact has only basic insulation to the winding, so it is kept on the mains side and never wired to V20 terminals.

S1 RUN, latching → DI1 (8) ON/OFF1
K1 NO, closed = motor OK → DI2 (9) ext. trip
S2 RESET, push button → DI3 (10) fault ack.
DI4 (11) not used

NOTES

1. Motor is connected in DELTA for a 3×230 V drive: U = Black + Red, V = Blue + Orange, W = Brown + Grey (from winding polarity test, 13.11.2026).
2. Never place a switch or contactor between V20 output (U/V/W) and motor.
3. E-stop S0 removes the V20 supply via K0. V20 has no STO. After reset the fan does NOT restart until S1 RUN is switched off/on (P1210 = 1).
4. Normal stop: S1 RUN off → ramp-down 20 s.
5. Max output 50 Hz (P1082). Wrong rotation: swap two of U/V/W at the drive, power off.
6. RCD 30 mA type A (see Siemens FAQ 49232264) or type B is permitted (manual p. 17).
7. Motor cable: shielded 4G1.0 mm², shield clamped at both ends; ≤ 10 m for class C2.
8. Control wiring 0.5–1.5 mm². Internal 24 V supply max 50 mA; 10 V max 11 mA.
9. Optional fault lamp on DO2 (terminal 18 NO / 19 C), max 230 V AC 0.5 A.
10. Before first run: insulation test 500 V, PE continuity, thermal trip test (F85), E-stop test. Full parameter list: workbook sheet 'V20 Wiring'.

Wire colours: L · N · Motor phase cable · Control · Thermal contact · PE

Drawn 13.11.2026 from bench measurements (Resistance + Polarity sheets) and SINAMICS V20 Operating Instructions A5E34559884. Status: DRAFT – not approved for installation.